

Backtesting & Statistics

Forward-Testing & Proving an Edge

Explained so simply a 10-year-old gets it

A flight simulator for traders — crash safely on old data, prove the plane works, and only then fly for real. Turning "I think" into measured evidence.

Stock Market Mastery Roadmap • Professional Level

Educational only — not investment, trading, or tax advice.

You've built a system (Chapter 21). Now, **before risking real money**, you test it against history and paper trades. **Backtesting** is like a **flight simulator for traders** — crash safely on old data, learn, and only fly for real once you've proven the plane works. Done right, it turns "*I think this works*" into **measured evidence**.

Backtesting logic: testing on the past, honestly

A backtest replays your rules over historical data. The trick is doing it **without cheating yourself**:

- **In-sample vs out-of-sample** — build the system on **one** slice of data, then test on a **fresh, untouched** slice. The second slice is the honest exam.
- **Walk-forward** — repeatedly build on the past and test on the next chunk, marching through time — closer to real life.
- **Look-ahead bias** 🧙 — accidentally using info you **couldn't have known yet** (e.g., today's close to decide today's open). A silent cheat that inflates results.
- **Survivorship bias** — testing only on companies that **survived**, ignoring the ones that died — makes any strategy look better than it was.

The metrics that actually matter

Metric	What it tells you
Expectancy	average profit per trade (in R)
Win rate	how often you win (alone, it's misleading!)
Avg win / avg loss	the size of wins vs losses
Profit factor	gross profit ÷ gross loss (>1 = profitable)
Max drawdown	the worst peak-to-bottom fall — the pain test
Sharpe / Sortino	return per unit of risk
CAGR	smoothed yearly growth rate
Recovery time	how long to climb back from a drawdown

MAX DRAWDOWN IS THE MAKE-OR-BREAK NUMBER

A system with amazing returns but a **60% drawdown** is useless — because **you'll quit (or blow up) before it recovers**. Max drawdown decides whether you can actually **stay in a system, emotionally and financially**. Always ask "could I sit through the worst stretch?" *before* the returns dazzle you.

The overfitting trap

CURVE-FITTING: MEMORISING INSTEAD OF LEARNING

Add enough knobs and you can make a strategy look **perfect on the past** — but it has just **memorised old noise**, not learned a real edge. It then **falls apart live**. Guard against it: use **few parameters**, and always confirm on **out-of-sample** data. **Out-of-sample survival is the real test of an edge** — nothing else counts.

Forward-testing: the dress rehearsal

A backtest lies a little (the past is tidy; the future is messy). So before real size:

- **Paper-trade** the system **live** for a while — same rules, no money — to catch real-world friction.
- Then trade **small real size** — real emotions show up only with real money.
- Only **scale up** once it survives all three stages. Rushing this is how good systems get abandoned for bad reasons.

Tools (from simple to serious)

- **Spreadsheet** 📊 — perfect for your **first** backtests; forces you to understand every step.

- **TradingView / Pine Script**  — a built-in strategy tester for chart-based rules.
- **Python**  (pandas, backtrader, vectorbt) — for the ambitious; reusable, powerful, and precise.

COSTS & SLIPPAGE: WHERE BACKTESTS LIE MOST

A backtest that ignores **brokerage, taxes, and slippage** (getting a worse price than expected) is a **fairy tale**. Real trading skims a little off every trade. Model these costs honestly — the gap between a clean backtest and messy reality is exactly this friction.



Key ideas to remember

- A backtest is a hypothesis, not a promise. Costs, slippage, and psychology **degrade** live results — always.
- Max drawdown decides whether you can actually stay in a system — emotionally *and* financially.
- Out-of-sample survival is the real test of an edge — in-sample brilliance means nothing on its own.

PRACTICAL EXERCISES

-  Backtest one rule in a spreadsheet on 100+ historical signals with realistic costs; compute the full metric set.
-  Split your data: build on the **first half**, validate on the **untouched second half**. If it dies out-of-sample, discard it.
-  (Optional) Reproduce the same test in **Pine Script or Python** to build reusable tooling.

REAL-WORLD APPLICATION

-  Given your software background, a small Python backtesting harness is a durable competitive advantage. It turns "*I think this works*" into **measured evidence** — and once built, you can test every new idea in minutes. That's an edge most retail traders never build.

RULE OF THUMB: BUILD WITH MARGIN TO SPARE

Assume your **live edge is meaningfully worse** than your backtest, once costs, slippage, and emotion take their cut. So **build systems with room to spare** — if it's only *barely* profitable on paper, it's probably a **loser** in real life. Aim for a comfortable cushion, not a razor-thin one.

ONE-LINE SUMMARY

Backtesting is a **flight simulator**: replay rules on history **honestly** (out-of-sample, no look-ahead/survivorship bias), judge with the **full metric set** (especially **max drawdown**), and beware **overfitting**. Then **forward-test** (paper → small → scale), **model costs & slippage**, and remember a backtest is a **hypothesis, not a promise** — build with margin to spare.